Genetic Algorithm-Based Optimization of PID Controller for a Coupled Tank System

Prepared as a part of an extracurricular activity beyond the prescribed curriculum, to enhance knowledge of genetic algorithm-based PID controller optimisation for a coupled tank system.

PREPARED BY

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01 Introduction

Liquid-level control is a common requirement in process plants, and a coupled tank system is a laboratory-scale model of cascaded storage vessels. The control action reaches the controlled level, the second-tank level (Tank 2 Level), through the dynamics of both tanks, so the loop shows lag and a tendency to overshoot [5]. The behaviour of a PID controller on such a loop depends strongly on its three gains [1].

This project tunes the PID gains of a Simulink coupled tank model (coupled_tank_model) with a Genetic Algorithm (GA) [3], [4] that minimises the Integral of Squared Error (ISE), and compares the result with a conventional PID controller (Kp = 2, Ki = 0.5, Kd = 0.1). All results come from MATLAB and Simulink simulation.

Scope note: Two GA gain sets appear in this report. The final optimisation run gave Kp = 6.107140, Ki = 0.192609 and Kd = 4.109572 (best ISE 1.04593208). The performance comparison used an earlier run (Kp = 6.246747, Ki = 0.279668, Kd = 4.795880). They come from different runs and are kept separate. Results are simulated and are not validated on a physical coupled tank.

02 Objectives

- To simulate a closed-loop PID level-control system for a coupled tank model in MATLAB and Simulink.
- To formulate PID tuning as an optimisation problem with decision variables Kp, Ki and Kd and the ISE as the fitness function.
- To implement the optimisation with the MATLAB Genetic Algorithm function *ga*.
- To compare conventional and GA-tuned PID using ISE, peak level, peak time, overshoot, steady-state error and settling time.
- To identify the trade-offs and the items that must be verified before the results are relied upon.

03 Software and Tools Used

No.	Tool / Block	Purpose
1	MATLAB	Scripts for optimisation, simulation runs, metric calculation and plotting
2	Simulink (model: coupled_tank_model)	Closed-loop simulation of the coupled tank and PID controller; stop time 30 s
3	ga (Global Optimization Toolbox)	Genetic Algorithm search for [Kp, Ki, Kd] [6]
4	PID(s) block and Saturation block	Controller and actuator limit in the loop
5	MATLAB Function block tank_dynamics, two integrators (1/s)	Computes dh1, dh2 from qin, h1, h2 and integrates to Tank 1 and Tank 2 Level
6	Logging block out.h2_data, mux and scope	Records Tank 2 level (h2_data) for analysis and displays both levels

MATLAB release and toolbox versions were not supplied: [insert release]. Levels are dimensionless model values; ISE has units (level units)²·s.

04 Theory

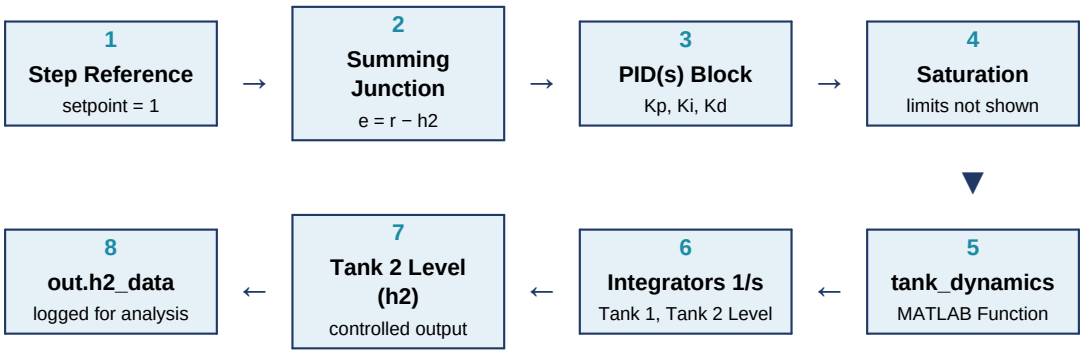
Item	Governing relation	Remark
PID control law	$u(t) = Kp\, e(t) + Ki \int e(\tau)d\tau + Kd\, de(t)/dt$	Kp, Ki, Kd = proportional, integral, derivative gains [1]
Transfer function	$C(s) = Kp + Ki/s + Kd\, s$	Ideal form; derivative filter of the project block to be confirmed
Error	$e(t) = r(t) - h2(t)$	r = setpoint (1); h2 = Tank 2 level
Fitness function	$ISE = \int e(t)^2\, dt,\ 0 \le t \le 30$ s	Penalises large errors strongly and small errors weakly
Overshoot	$\%OS = (y_{peak} - r) / r \times 100$	Used for the overshoot metric
ISE reduction	$(ISE_{conv} - ISE_{GA}) / ISE_{conv} \times 100$	Used to compare controllers
Decision vector	$x = [Kp, Ki, Kd]$	Chromosome of three real genes

Genetic Algorithm Elements

- **Population and fitness:** each candidate gain vector is simulated and scored by its ISE; smaller is better.
- **Selection:** parents are chosen in favour of better fitness (stochastic uniform).
- **Crossover and mutation:** scattered crossover recombines parents; adaptive feasible mutation adds variation within the bounds.
- **Termination:** a generation limit or a stall of the best fitness. The search is stochastic, so different runs give different gains [8].

Equations are standard and are not claimed to be the equations inside tank_dynamics, whose contents and physical parameters were not supplied.

05 Block Diagram



▲ h2 is fed back to the summing junction (negative input) ▲

Figure 1: Signal flow of the control loop in coupled_tank_model.

The reference is compared with the measured Tank 2 level, and the error drives the PID block. The saturation block limits the controller output applied as inflow qin to the tank_dynamics block. Its outputs dh1 and dh2 are integrated into the two tank levels, and h2 is both fed back and logged as out.h2_data.

06 Flowchart

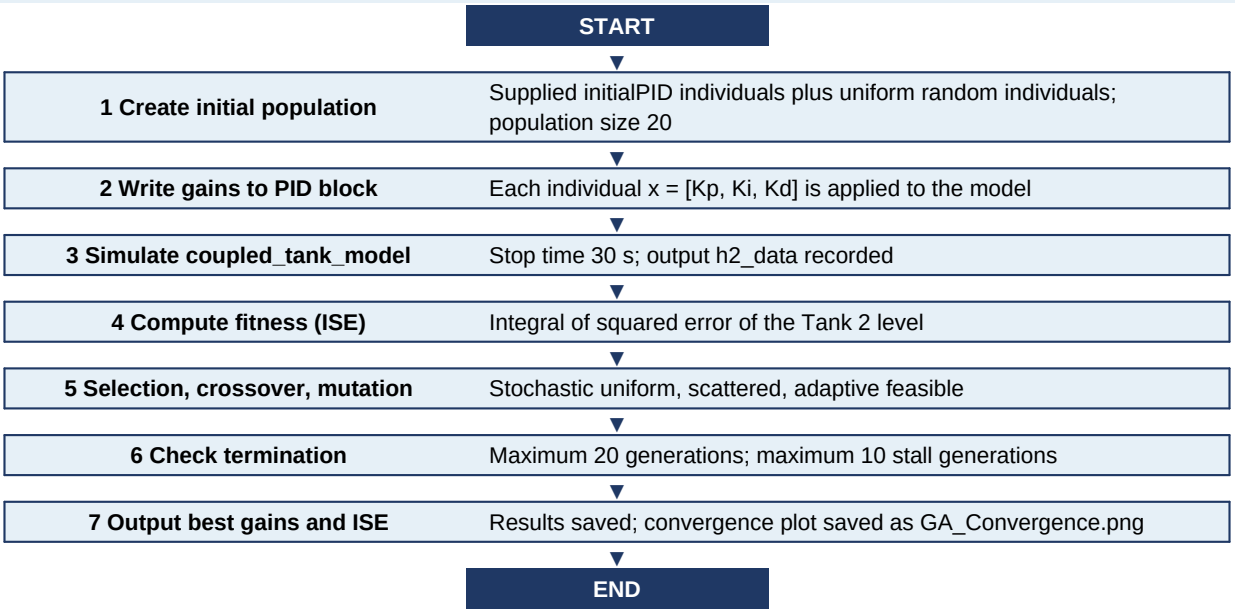


Figure 2: Flowchart of the GA-based PID optimisation.

07 Simulink Model Design

The control loop was built in Simulink as the model coupled_tank_model (Figure 3). A step source gives the setpoint, the PID(s) block and a saturation block drive the tank_dynamics MATLAB Function block, and two integrators give Tank 1 Level and Tank 2 Level.

Simulink Model

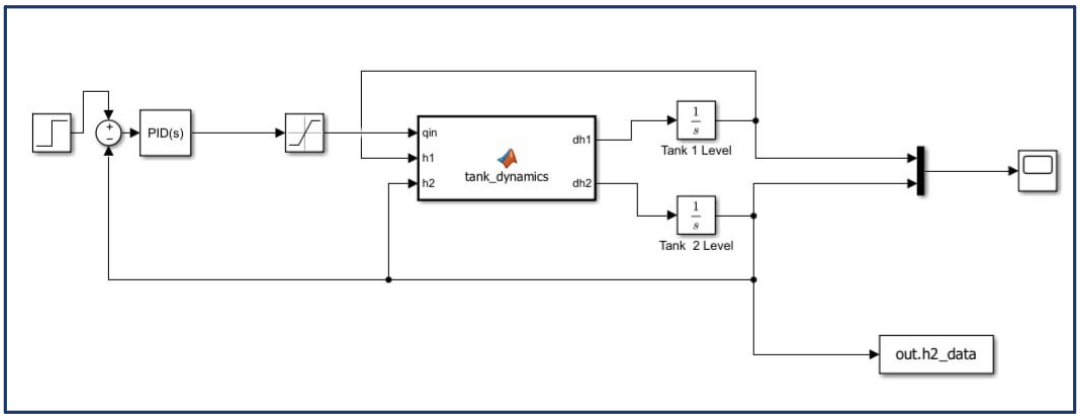


Figure 3: Simulink model of the coupled tank system with PID controller.

Model Stages and Values

Stage	Main blocks and values	Function
Reference	Step block; setpoint = 1	Desired Tank 2 level; step time not shown
Error	Summing junction (+ reference, − h2)	Forms e(t) = r − h2
Controller	PID(s) block; gains from table below	Computes the control signal
Actuator	Saturation block	Limits the inflow command; limits not shown
Plant	tank_dynamics (inputs qin, h1, h2; outputs dh1, dh2); two 1/s integrators	Produces Tank 1 and Tank 2 levels
Output	out.h2_data; mux and scope	Logged Tank 2 level (h2_data); both levels displayed
Simulation	coupled_tank_model; stop time 30 s	Same model and stop time for both controllers

Design Review and Flagged Items

The supplied material was checked for consistency. The items below should be resolved or verified before the results are treated as final.

Priority	Finding	Recommended action
High	**Two GA gain sets.** The comparison used an earlier run; the final-run gains (best ISE 1.04593208) were not separately simulated against the conventional controller.	Re-simulate with the final gains, compute all six metrics and update Section 09.
High	Steady-state error is higher for the GA-tuned comparison controller (0.005060 against 0.000293).	Add a steady-state term or ITAE/ITSE to the objective, or test a longer simulation.
Medium	GA ended at the generation limit (20 generations, exit flag 0) with a small population of 20; the curve was nearly flat but not proven converged.	Run more generations and several trials.
Medium	Bounds and initialPID contents were not supplied; the GA source code was not supplied.	Insert bounds, initialPID and the scripts as an appendix.
Medium	**Metric definitions.** Settling-time band and steady-state-error instant used by the script were not supplied (±2% is assumed to be consistent with the plots).	Confirm from the comparison script.
Medium	tank_dynamics equations and physical parameters were not supplied, so no plant model is stated.	Document the equations and parameters.
Check	Figure 5 is titled only "Tank 2 Liquid-Level Response"; it is identified as the conventional PID response because it matches the conventional curve in Figure 7.	Confirm and retitle the plot.
Check	Saturation limits, step time, solver settings, PID block form and derivative filter are not visible.	Insert from the model.

Priority	Finding	Recommended action
Note	Results are from simulation only; behaviour on physical equipment is not claimed.	Validate on hardware before industrial use.

08 Working Principle and GA Optimization

1. Each candidate x = [Kp, Ki, Kd] is applied to the PID block and coupled_tank_model is simulated for 30 s.
2. The ISE of the Tank 2 level error is returned to the GA as the fitness value.
3. The GA selects, recombines and mutates the candidates over the generations to reduce the best ISE.
4. The run stops at the generation limit; the best gains and ISE are saved.

GA Settings (final reported run)

Option	Setting	Purpose
PopulationSize	20	Candidate gain vectors evaluated per generation
MaxGenerations	20	Upper limit on generations
MaxStallGenerations	10	Stops the run if the best fitness stalls for 10 generations
CreationFcn	@gacreationuniform	Random individuals sampled uniformly within bounds
CrossoverFcn	@crossoverscattered	Each gene taken from one of two parents by a random mask
SelectionFcn	@selectionstochunif	Stochastic uniform selection of parents
MutationFcn	@mutationadaptfeasible	Adaptive mutation that keeps offspring within bounds
InitialPopulationMatrix	initialPID	Starting gain vectors for the first generation
Plot function	Disabled (later version)	Avoids plot-related interruption during the run

Other options not shown in the supplied code are assumed to be MATLAB defaults. The run completed 20 generations; earlier runs used different configurations, so this is not claimed for every run.

Final Reported GA Optimization Results

Quantity	Value
Optimised proportional gain, Kp	6.107140
Optimised integral gain, Ki	0.192609
Optimised derivative gain, Kd	4.109572
Best ISE reported by GA	1.04593208
Generations completed	20
Exit flag	0 (maximum generations reached)

Convergence of the Final Run

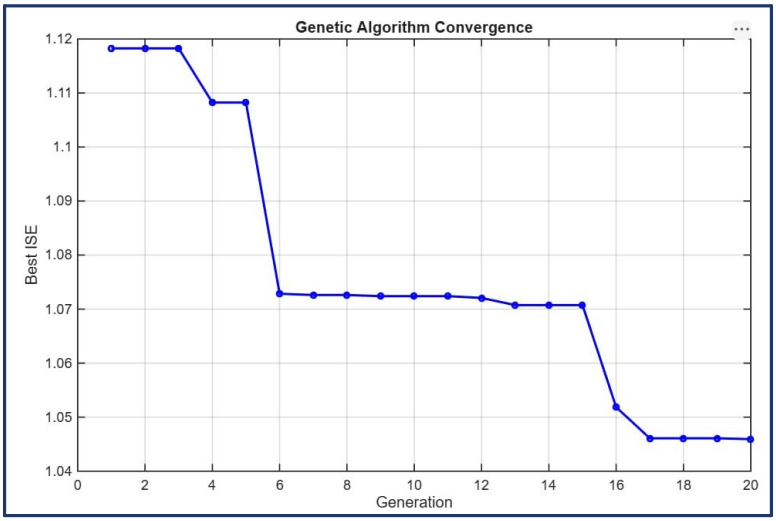


Figure 4: Genetic Algorithm convergence graph (best ISE versus generation, GA_Convergence.png).

Values are read approximately from the graph (grid 0.01). The best ISE is about 1.12 in generations 1–3, about 1.11 in generations 4–5, and falls to about 1.07 at generation 6. It changes little until generation 15, drops to about 1.05 at generation 16 and to about 1.046 at generation 17, then stays nearly flat to generation 20. The final value agrees, within graph resolution, with the reported best ISE of 1.04593208. Because the run ended at the generation limit, full convergence cannot be concluded.

Gain Sets Used in This Report

Controller / Gain set	Kp	Ki	Kd	Used for
Conventional PID	2	0.5	0.1	Baseline in Section 09
GA-tuned (comparison-script set)	6.246747	0.279668	4.795880	Comparison table and graphs
GA-tuned (final optimisation run)	6.107140	0.192609	4.109572	Final optimisation result

Important: The comparison table and graphs in Section 09 were produced with the comparison-script gain set, not the final optimisation gains. The two sets come from different optimisation runs.

09 Simulation Results

All results use coupled_tank_model, a stop time of 30 s, the output h2_data and a setpoint of 1. The GA-tuned results use the comparison-script gains (Kp = 6.246747, Ki = 0.279668, Kd = 4.795880).

32.85%	39.95% → 4.42%	14.496 s → 5.901 s	0.000293 → 0.005060
ISE reduction	Overshoot	Settling time	Steady-state error (increased)

Conventional PID Response

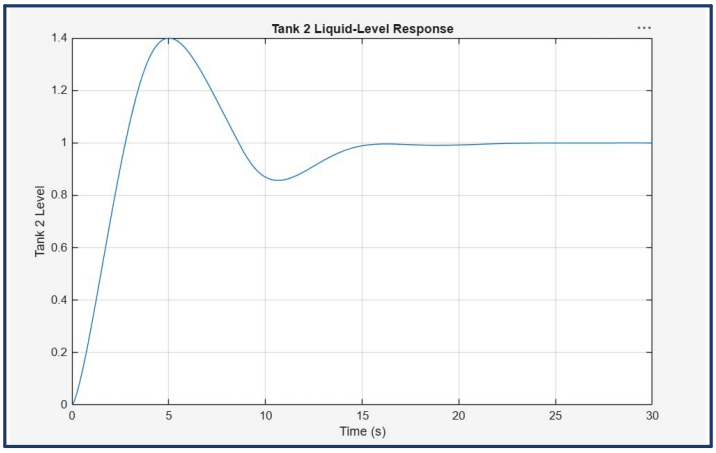


Figure 5: Tank 2 level response, conventional PID (Kp = 2, Ki = 0.5, Kd = 0.1).

The level peaks at about 1.4 near 5 s, undershoots to about 0.86 near 10.5 s and settles near 1 after roughly 15 s: an under-damped response.

GA-Tuned PID Response

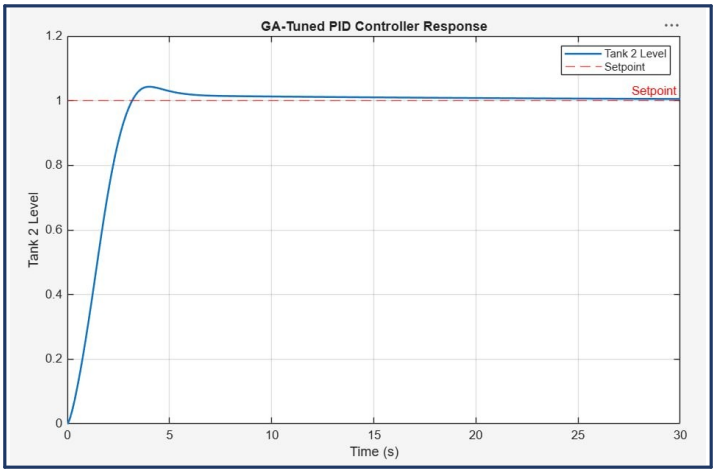


Figure 6: Tank 2 level response, GA-tuned PID (comparison-script gains), with the setpoint.

The response peaks slightly above 1 at about 4 s and then decays smoothly towards the setpoint without undershoot. At 30 s it is still marginally above the setpoint.

Response Comparison

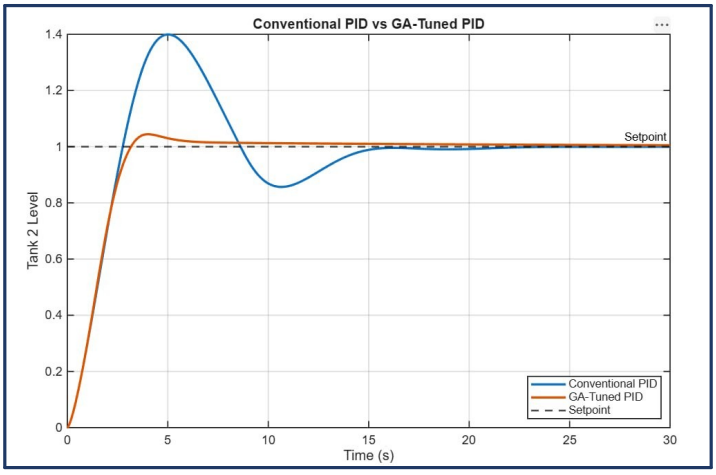


Figure 7: Conventional PID versus GA-tuned PID Tank 2 level response with the setpoint.

The two curves nearly coincide for about 2 s. The conventional response then overshoots to about 1.4 and oscillates, whereas the GA-tuned response levels off near 1.04. The GA-tuned curve remains slightly above the setpoint at the

end of the interval.

Performance Comparison

Metric	Conventional PID	GA-Tuned PID
Kp, Ki, Kd	2, 0.5, 0.1	6.246747, 0.279668, 4.795880
ISE ((level units)²·s)	1.561873	1.048785
Peak Level (level units)	1.399459	1.044224
Peak Time (s)	5.008	3.995
Overshoot (%)	39.95	4.42
Steady-State Error (level units)	0.000293	0.005060
Settling Time (s)	14.496	5.901

Derived Changes (calculated from the reported values)

Metric	GA-tuned relative to conventional
ISE	Reduced by 0.513088 (32.85%)
Peak level	Reduced by 0.355235
Peak time	Earlier by 1.013 s (20.23%)
Overshoot	Reduced by 35.53 percentage points
Settling time	Reduced by 8.595 s (59.29%)
Steady-state error	Increased by 0.004767 (about 17.27 times)

ISE reduction: (1.561873 − 1.048785) / 1.561873 × 100 = 0.513088 / 1.561873 × 100 ≈ 32.85%. This applies to the comparison-script gains only; no reduction is calculated for the final-run gains.

Interpretation

- The GA-tuned comparison controller gives lower ISE, peak level, peak time, overshoot and settling time.
- It does not improve every metric: steady-state error rises from 0.000293 to 0.005060 (about 17.27 times; about 0.5% of the setpoint).
- Likely reasons (interpretation, not separately tested): a smaller integral gain (0.279668 against 0.5), a slow approach to the setpoint from above within the 30 s window, and the low weight ISE gives to a small persistent error.
- Performance should be judged with several metrics, not ISE alone.

Note: These results are from simulation. They do not guarantee identical performance on physical equipment, where noise, actuator limits and disturbances apply. The large derivative gain may amplify measurement noise.

10 Advantages and Applications

Advantages
- Tuning criterion (ISE) is stated and optimised directly
- No gradient information or linearisation needed
- Automated and repeatable in MATLAB
- Much lower overshoot and shorter settling time in simulation
- Same procedure can be reused with other objectives

Applications
- Level control of tanks and storage vessels (after validation)
- Simulation-based tuning of PID loops
- Comparing tuning methods on a benchmark process
- Teaching PID tuning and optimisation
-

11 Limitations and Future Scope

Limitations

- Simulation only; no hardware validation
- Small GA population (20) and 20 generations
- Single objective (ISE), one setpoint and one stop time
- Results vary between GA runs; no repeated-trial statistics
- Final-run gains not separately compared with the conventional controller
- Plant equations and parameters not reported

Future Scope (not yet performed)

- Larger population, more generations and multiple trials
- Comparison with Particle Swarm Optimization and other metaheuristics
- Other objectives such as ITAE and ITSE, or a steady-state term
- Testing under different setpoints and disturbances
- Validation on a physical coupled tank setup

12 Conclusion

A GA-based procedure for tuning a PID controller was implemented in MATLAB and Simulink for the Tank 2 level of a coupled tank model, with ISE as the fitness function. The final optimisation run (population 20, 20 generations, exit flag 0) gave Kp = 6.107140, Ki = 0.192609 and Kd = 4.109572 with a best ISE of 1.04593208.

In the separate comparison over 30 s, the GA-tuned gains Kp = 6.246747, Ki = 0.279668 and Kd = 4.795880 reduced the ISE by 32.85%, the overshoot from 39.95% to 4.42% and the settling time from 14.496 s to 5.901 s relative to the conventional controller, but increased the steady-state error from 0.000293 to 0.005060. The flagged items in Section 07, mainly re-simulation with the final gains and hardware validation, should be resolved before the results are treated as final.

13 References

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Edition, year and publisher details are to be added as per the copies used.